



Discrete Optimization



The wildfire suppression problem with multiple types of resources

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ABSTRACT

The frequency and impact of wildfires have considerably increased in the past decade, due to the extreme weather conditions as well as the increased population density. The aim of this study is to introduce, model, and solve a wildfire suppression problem that involves multiple types of fire suppression resources and their operational characteristics. Two integer programming (IP) formulations, a basic IP and its reformulation with combinatorial Benders' cuts, are presented. The performances of the proposed formulations are evaluated on a set of randomly generated instances. The results indicate that the proposed formulations are able to obtain high quality upper and lower bounds. Extensive numerical experiments are performed to analyse the effects of several operational constraints on the computational performance of the models. A case study arising in Yatağan district of Muğla province of Türkiye is presented.

1. Introduction

Due to global warming and population growth, the frequency of wildfires has increased significantly, resulting in many adverse effects on the environment, economy, and society. Ignited by the hot and dry weather conditions in the summer of 2009, the wildfires displaced at least 1500 people in Spain and caused the loss of 4000 hectares of forest area (BBC News, 2009). Wildfires caused by severe drought in Australia in 2019 damaged more than 2000 homes and 200 facilities, and killed at least 27 people (Newsweek, 2020). Türkiye has been significantly affected by the wildfires, with 139,503 hectares of forest land burned in 2021 (Orman Genel Müdürlüğü, 2022). The majority of wildfires broke out in the first week of August 2021. In the same week, Greece was struggling with wildfires as well, due to the severe heatwave experienced in the Mediterranean region at that time (European Commission, 2022). It has been demonstrated in Turco et al. (2017) that there is a significant relationship between the wildfires and the droughts in Mediterranean Europe. This outcome supports the idea that drought to be experienced due to the global climate change will increase the frequency and severity of the wildfires in regions located in the mid-latitudes such as the Mediterranean region. Moreover, the wildfires are expected to increasingly affect countries located further north, such as the UK. In 2019, 137 wildfires have been reported in the UK, the highest number in the country's history (European Commission, 2020).

Response to a wildfire is a complex process, and the decisions on how to tackle the fire can significantly influence the damage to the

forest, as well as properties and infrastructures. Many types of fire suppression resources can be deployed to effectively fight a wildfire, but each resource has different operational restrictions and impact on suppressing the fire. To the best of our knowledge, there is no study that optimises wildfire suppression plans by considering different types of resources with distinct operational characteristics and impacts on the spread of the wildfire.

The purpose of this study is to close this research gap by introducing the Wildfire Suppression Problem (WSP), where different types of resources, their operational constraints and the direct and indirect impacts on wildfire spread are taken into consideration. The objective is to minimise the damage caused by the wildfire, allowing the decision makers to prioritise infrastructure, inhabited areas, and fauna and flora under the threat of extinction.

Therefore, our aim is to provide optimisation models that can:

- Provide precise and coordinated wildfire response plans in which the practical characteristics of the resources and their direct and indirect impacts on wildfire spread are taken into account;
- Allow for scenario analysis so as to quantify the importance of alternative resources;
- Support the creation of more reliable and effective fire suppression plans (see The Scottish Government, 2013).

Our problem definition has been tailored based on a set of interviews with the Muğla Regional Directorate of Forestry in Türkiye, and

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informative discussions with the UK Forestry commission, Natural Resources Wales and Forestry England. These organisations informed our modelling decisions and highlighted the importance of the practical characteristics of different types of suppression resources. We have developed a polynomial-size basic integer programming (IP) formulation for the WSP.

We also present a reformulation and associated solution method based on *combinatorial Benders' cuts* (Codato & Fischetti, 2006). This method is essentially based on working with a smaller formulation that allows infeasible solutions, which are in turn declared infeasible as they are encountered. It has been used for the models regarding rebalancing of shared bicycle systems (Erdoğan et al., 2015), network interdiction (Smith & Song, 2020) and more recently, crowdshipping (Boysen et al., 2022). Based on this method, we provide a reformulation of WSP with an exponential number of feasibility constraints, where the decision variables employed to define specific assignments of firefighting resources to extinguishing operations are removed. Every time a feasible solution to the reformulation is found, a procedure checks if a feasible firefighting resource assignment exists for the extinguishing plan contained in the solution. If this is not the case, a feasibility constraint in the form of combinatorial Benders' cuts is used to separate the solution found to be infeasible.

Both formulations are strengthened by a pre-processing technique and valid inequalities. The performances of the formulations have been compared on a set of randomly generated benchmark instances. The computational results underline the superiority of the exponential-size formulation over the basic one. In addition, we have made several computational experiments to assess the impact of various operational constraints on the success of suppression plans. Finally, we present a case study related to an application of our modelling approach to a wildfire suppression case in Yatağan district of Muğla province, Türkiye.

The rest of the paper is organised as follows: In Section 2, the studies that utilise Operational Research (OR) methodologies to wildfire suppression planning are reviewed. Section 3 provides the description of the problem and our modelling approach. Section 4 presents our mathematical formulations developed for the WSP. In Section 5, we provide the results of computational analysis on the randomly generated instances. Section 6 contains a case study based on data from Yatağan district of Muğla province, Türkiye. Finally, Section 7 presents conclusive remarks and possible future research directions.

2. Literature review

There is a substantial body of research on the applications of OR methodologies to wildfire management problems such as wildfire vulnerability assessment (Rashidi et al., 2017), fuel management (Bhuiyan et al., 2019; Liberatore et al., 2021; Matsypura et al., 2018), and wildfire suppression (Duff & Tolhurst, 2015). We refer the interested reader to the comprehensive review of Minas et al. (2012) on the applications of OR to wildfire management problems. The rest of this section is devoted to the review of articles related to OR applications to wildfire suppression.

In a recent review, Granda et al. (2023) have classified the wildfire suppression models into four groups, namely, fire-line based, fire points based, grid based, and potential wildfire operation delineations (POD) based models. Fire-line based models focus on the containment of fire perimeter by a control line (Donovan & Rideout, 2003; Hu & Ntamo, 2009; Rashidi et al., 2018a; Rodríguez-Veiga, Ginzo-Villamayor, & Casas-Méndez, 2018). Fire points based models consider multiple fire events as a set of demand points, and the fire is contained when all the fire points have been visited by the resources (Bodaghi et al., 2020; Rodríguez-Veiga, Gómez-Costa, et al., 2018; Shahparvari et al., 2021). In the grid based models, forest land is represented by a grid consisting of cells which may have different characteristics (Alvelos, 2018; Belval et al., 2015; Mendes & e Alvelos, 2023; Wei et al., 2011; Zhou &

Erdoğan, 2019). POD based models discretise forest land by using PODs which are predefined land divisions with similar characteristics (Wei et al., 2021, 2018, 2019). A simplifying assumption of the fire-line based models is that the fire is contained when the perimeter of the control line is larger than that of the fire. Similarly, fire points based models assume that the fire is contained when all the fire points have been visited. These assumptions may be appropriate in initial attack modelling. However, they are insufficient to model the extended attack in cases of large wildfires in which suppression efforts have an impact on the fire spread. On the other hand, grid based models are capable of covering the interaction between the suppression efforts and fire spread. The defined fire spread behaviour allows modelling without the need for a stand alone fire simulator.

The grid based modelling approaches are theoretically related with the Firefighter Problem (FP) (Hartnell, 1995). The FP considers a raster grid in which each cell can be burnt, protected, or untouched. At each given time instant, the resources to protect some of the cells have to be assigned, and the fire spreads over to the cells that are adjacent to the burnt cells, unless they are already burnt or protected. Finbow and MacGillivray (2009) provide a more comprehensive review of the FP and its extensions. García-Martínez et al. (2015) developed an IP model and heuristic algorithms for the FP. Ramos et al. (2020) proposed improved IP formulations and a matheuristic algorithm for the FP. Zambon et al. (2018) studied the Geometric FP (GFP), which consists of locating barriers (i.e., modelled as chords of a polygon) to block the spread of a fire and limit its damage. The authors developed an IP model for the GFP, and pre-processing techniques to reduce the problem size. In a subsequent work, Zambon et al. (2019) generalised the GFP by considering the paths for the barrier construction. Although the FP is studied in depth in the literature, most of these studies do not address the practical aspects of wildfire suppression (Granda et al., 2023).

There exist a number of studies which employ grid based modelling approaches to address the practical aspects of wildfire suppression. In these studies, the impact of suppression efforts on fire spread takes place in the models as either a delay in the fire arrival times or a change in the state of the cells. The studies in the first group model the fire spread by utilising the fire arrival times. If a resource is deployed in a cell, the fire spread from that cell is delayed by a prespecified amount. Accordingly, the fire arrival times of the related cells are updated. In this context, Wei et al. (2011) developed a MILP model for the suppression of large wildfires. The model is defined on a raster landscape where the effects of fuel and topography on fire spread can be incorporated into the model. The model assumes an elliptical fire spread and calculates the earliest arrival times of the fire to each cell. If a resource is deployed in a cell, the fire spread from that cell is delayed by a specified amount. Alvelos (2018) proposed four MILP models, which differ in their objectives, by considering the protection of specific areas, total burnt area, and fire containment. The earliest fire arrival times are incorporated into the models by using shortest path tree concepts. Moreover, the authors introduced a new definition for fire containment. They defined the fire containment as the time at which there are no more new fire ignitions. Mendes and e Alvelos (2023) developed an iterated local search algorithm for the problem described by Alvelos (2018) to obtain high-quality solutions in reasonable computation times.

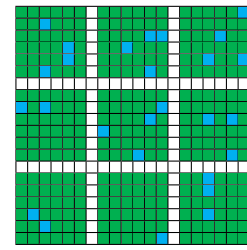
As an alternative approach, the fire spread can be modelled with the states of the cells. In such models, the fire spreads from a cell in active fire to its adjacent cells. If a suppression resource is deployed in a cell, fire spread from that cell is stopped, which delays the fire arrival times to its adjacent cells consequently. In this regard, Belval et al. (2015) developed a MILP model for the resource allocation problem in wildfire suppression. They considered fire intensity on their spatial wildfire spread model on raster landscapes. The fire spreads from a burnt cell to its adjacent cells. If a suppression resource is deployed in a cell, the cell will not burn. Then, the authors extended the model with

multiple wind scenarios in Belval et al. (2016). The model is further generalised by Belval et al. (2019) to cover the characteristics of ground suppression activities. Zhou and Erdogan (2019) presented a two-stage stochastic goal programming model for the firefighting resource allocation and resident evacuation problem. The model considers multiple wildfire spread scenarios. The problem is defined on a raster landscape with variable cell sizes. The cell sizes depend on the fire spread rate in the relevant scenario. The fire spread is modelled spatially where the cells in the fire spread direction are marked as high-risk cells. When a high-risk cell is protected by the resources, it will not burn.

The majority of the models in the literature treats the suppression resources as identical and always available. In a few studies, the operational characteristics of suppression resources are incorporated into the models. Rodríguez-Veiga, Gómez-Costa, et al. (2018) developed a MILP model to allocate air resources to a set of firefronts by considering resting times and refuelling constraints. The main difference of their model from our model is that the wildfire spread is not considered. Shahparvari et al. (2021) proposed a MILP model for the coordination of aerial suppression resources. The model minimises the total completion time and total costs. This study differs from ours in that the wildfire spread is not incorporated into the model. Instead, the resource requirements of each period are prespecified. Rodríguez-Veiga, Ginzo-Villamayor, and Casas-Méndez (2018) considered the working shifts and resting times of fire brigades and aerial resources in their MILP model. As their model is a fire-line based model, it does not address the impact of suppression on fire spread and provide a detailed response plan for each resource. In contrast, our model incorporates both direct and indirect effects of suppression activities. Belval et al. (2019) developed a grid based model which addresses control line quality, connectivity of firefighter travel paths, and safety constraints in ground suppression activities. In this study, a firefighter team begins travelling from one of the predefined access points, and can travel to the adjacent cells. It is assumed that the teams cannot travel through a cell more than once. Our modelling approach does not require such an assumption. In our model, the ground resources can only access to the incident zone through forest roads. In Belval et al. (2019), the effect of firefighting is modelled by a control line which reduces the fire intensity. In our model, the direct and indirect effects of firefighting and firebreak construction activities are defined in more detail. Moreover, the safety distance is dynamically determined according to the fire intensity in Belval et al. (2019). In our case, we have defined safety distances for the resources by taking the wind direction into account. Consequently, the safe working zones are updated when the wind direction changes. Note that characteristics of the terrain, as the slope and the type of vegetation, can also be easily incorporated in our fire spread model.

To the best of our knowledge, there is no model that considers the coordination of multiple types of resources with their operational characteristics such as safety distance, availability of access to the firefront, and response time delays. Moreover, the effect of suppression activities on wildfire spread is modelled simplistically - a protection of a cell or a delay in the earliest fire arrival time. In this regard, the contributions of our study to the related literature are as follows:

- We consider different types of resources (firefighter teams, fire-break construction teams, and firefighting helicopters) and their operational characteristics. This allows for a refined suppression plan, given that the resources need to operate under different constraints and can achieve different outcomes in extinguishing and limiting the spread of the fire.
- The objective function of our model includes the damage caused by razing the forest land in addition to the burnt areas in overall damage calculation. Our model allows for prioritising infrastructure and areas of natural interest.
- Our model explicitly accounts for the availability of access to the firefront through road networks. Firefighter teams and firebreak construction teams can only reach parts of the forest that are covered by road networks, leaving only the possibility of aerial resources to protect the rest of the area.



■ Flammable ■ Nonflammable □ Forest road

Fig. 1. An example raster landscape. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

- Our model covers the direct and indirect effects of different types of resources on fire spread by defining a specified fire spread rate for each cell on raster grid.

3. Problem definition and optimisation model

Consider a raster landscape that is comprised of identical-sized square cells in a rectangular area. A 20×20 example is depicted in Fig. 1, where the green and blue cells represent the flammable and non-flammable areas, respectively. A wildfire can ignite in the flammable cells and spread on to them. Each flammable cell is associated with a value. Cells belonging to a wildlife protection area, a populated area, or an area close to critical infrastructure are particularly important. The nonflammable cells are associated with lakes, rocky areas, or fire breaks. The white cells denote the forest roads. The firefighter teams can access into the forest by these roads. A wildfire cannot burn the nonflammable and forest road cells. Therefore, these cells have no value. However, a wildfire can pass over them if its flame length is large enough.

In the following subsections, we initially describe the fire spread pattern. Then, the considered wildfire suppression tactics are explained. Finally, a proof of $\mathcal{NP}$ -hardness is provided for the considered WSP.

3.1. Fire spread pattern

The simplest fire spread pattern is circular in which the fire spreads from the ignition point at an equal rate in all directions. In the presence of wind, the fire spreads at a higher rate in its direction. In this case, the fire spread pattern resembles an ellipsoid (Anderson et al., 1982; Green et al., 1983). The elliptical fire spread pattern is illustrated in Fig. 2. In Fig. 2, Zones (a), (b), and (c) are the areas burnt by head fire, flank fire and back fire, respectively. The intersection point of all three zones is the ignition point.

In a few study, the elliptical fire spread pattern has been incorporated into grid based modelling approaches (Wei et al., 2011; Zhou & Erdogan, 2019). In this study, we adapt the elliptical fire spread pattern on a raster landscape as follows: Let T be the planning horizon which consists of equally spaced $t_{max} + 1$ time points, $T = \{0, \dots, t_{max}\}$. Let σ_h , σ_f , and σ_b be head fire, flank fire, and back fire spread rates, given in number of cells per time period, respectively. Assume that $\sigma_h > \sigma_f > \sigma_b$. For each time period between time points t and $t + 1$, there are four possible wind directions. In one time period, the firefront progresses in the wind direction by σ_h . In the direction which is orthogonal to the wind, the firefront advances by σ_f . Along the cells between the head fire and the flank fire directions, it grows by $\lceil \sigma_h/2 \rceil$ and $\lceil \sigma_f/2 \rceil$ in the head and flank fire directions, respectively. In the opposite direction of the wind, it spreads by σ_b . Through the cells between the flank and back fire directions, it expands by $\lceil \sigma_f/2 \rceil$ and $\lceil \sigma_b/2 \rceil$ in the flank fire and the back fire directions, respectively.

An example fire spread on the raster landscape is illustrated in Fig. 3. At time point $t = 0$, a fire ignites in the cell which is indicated in

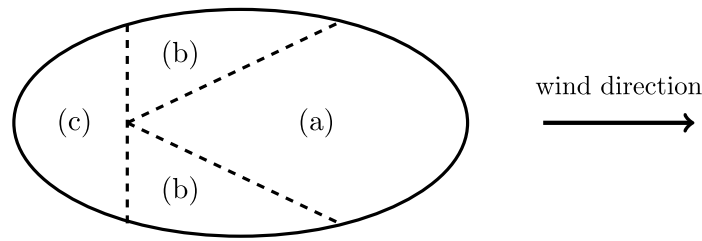


Fig. 2. The elliptical fire spread pattern (Wagner, 1969).

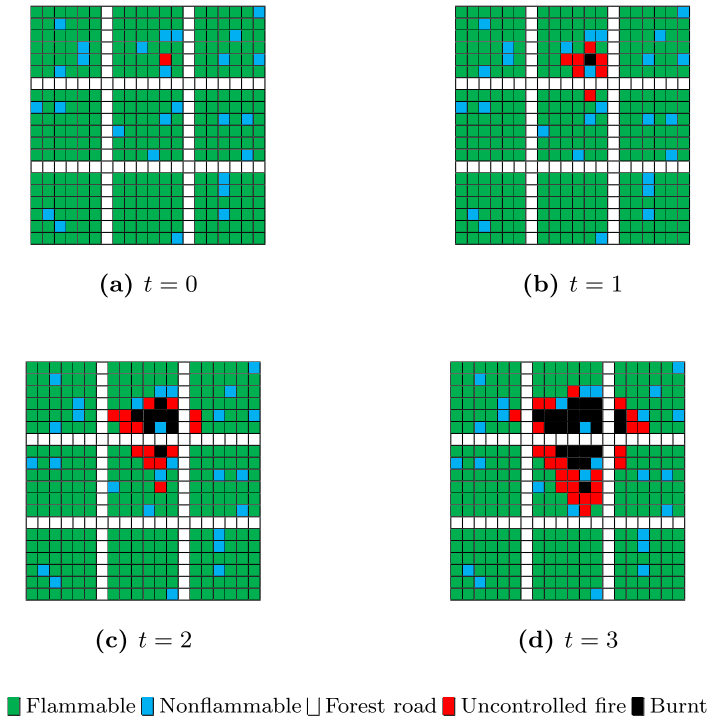


Fig. 3. An example of fire spread on a raster landscape. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

red (Fig. 3(a)). The active firefront is denoted by red colour, whereas the burnt cells are depicted in black. In this example, the wind direction is from above to below in all time points. The fire spread rates are $\sigma_h = 3$, $\sigma_f = 2$, and $\sigma_b = 1$. Therefore, the fire ignited at $t = 0$ spreads into the below three cells at the next time point (Fig. 3(b)). It progresses by two cells in the flanks and one cell in the back as shown in Fig. 3(b). The cells that have caught fire in previous time points are completely burnt and cannot catch fire again. This assumption is in line with the characteristics of fire spread and suppression efforts. The cells belonging to firefront spread fire to farther cells than the inner cells do. Therefore, suppression efforts target the firefront instead of the fire inner area. In this case, the inner area is left to burn. The fire can cross over forest roads, nonflammable cells and burnt cells, if its spread rate is sufficiently high. In Fig. 3(b), the head fire spreads on the forest road but the related forest road cell is not damaged in the next time point (Fig. 3(c)). Similarly, the head fire reaches a nonflammable cell in $t = 1$ but it cannot burn it. Furthermore, two adjacent nonflammable cells serve as a firebreak in $t = 2$, and the spread of the back fire is blocked in this direction (see Figs. 3(c) and 3(d)).

3.2. Wildfire suppression

The wildfire suppression tactics that we consider are ground and aerial firefighting, and firebreak construction. We have informed about

the implementation details of these tactics by the interviews conducted with Muğla Regional Directorate of Forestry, and developed a modelling approach on raster landscapes accordingly. The following subsections present the implementation of these tactics and the details of our modelling approach on raster landscapes.

3.2.1. Ground and aerial firefighting

The Firefighting Teams (FTs) are responsible for the ground firefighting. An FT consists of a wildland fire engine, a water tanker, and five firefighters. The wildland fire engines use water and firefighting foams for fire suppression. The fire engines and water tankers must travel fully loaded. The water tanker can be refilled from natural water resources such as sea, rivers, and lakes, as well as artificial lakes. The fire engines can be replenished by the water tankers without leaving its location. The fire engines are 4×4 vehicles that are able to travel through the forest roads. They cannot leave the forest roads and the firefighters access the firefront by laying hoses into the forest. Therefore, the length of the hoses constrains the extinguishing range of an FT. The fire engines should be at a safety distance from the firefront. The safety distance depends on the current wind direction. The fire engines cannot be directly in front of the head fire. The safety distance from flank fire is larger than that from the back fire.

Consider the wildfire case illustrated in Fig. 3(c). Suppose that the extinguishing range is seven cells and the safety distance is two cells.

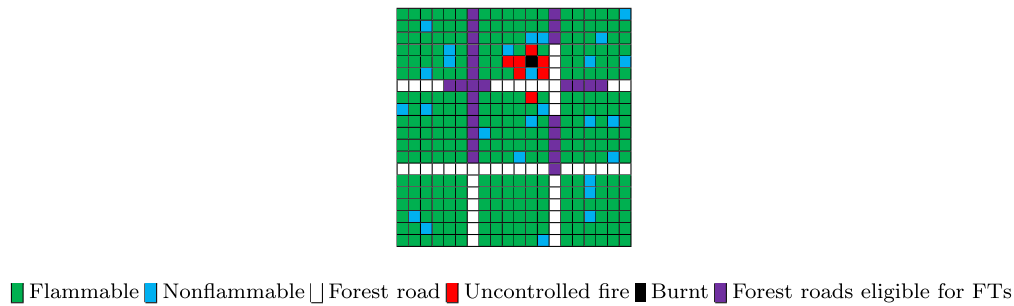


Fig. 4. The forest road cells in which a fire engine can be placed in $t = 1$.

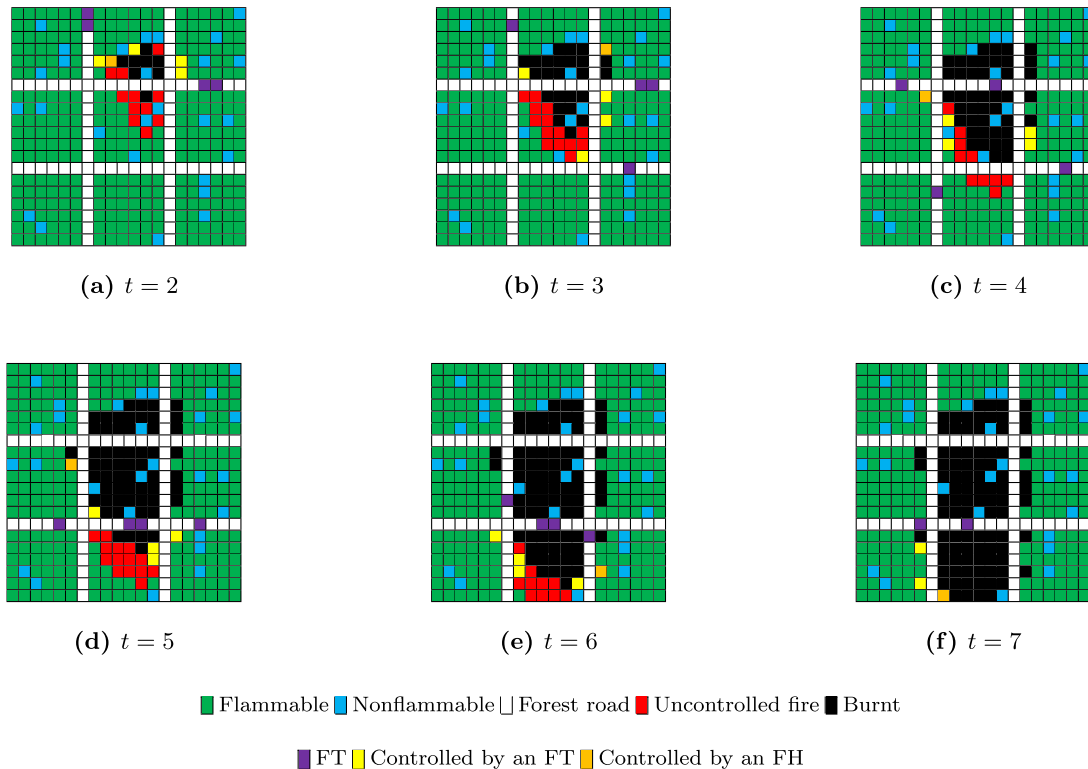


Fig. 5. An example to firefighting operations.

In Fig. 4, the forest road cells in which a fire engine can be located are presented in purple colour.

For aerial firefighting, the Firefighting Helicopters (FHs) are the primary resources. The FHs are more advantageous in approaching the firefront since they do not have the safety constraints of ground firefighting. An FH drops all of its load of water in one turn. It then needs to be replenished from the sea, lakes, or artificial lakes.

The firefighting tactics are employed in both initial attacks and escaped wildfires. In initial attack, the aim of firefighting is to put out the fire by extinguishing it. On the other hand, it is challenging to suppress an escaped wildfire by employing the firefighting resources solely. In this case, the firefighting resources attempt to delay the firefront progress. In both cases, as a result of firefighting, the extinguished area is damaged but the fire cannot spread from it. Fig. 5 shows an example to our modelling approach for firefighting operations on the raster landscape. In this example, there are four FTs and an FH, each of which is capable of extinguishing a single cell in one time period. The range of an FT is seven cells. The safety distance for fire engines is two cells. Suppose that firefighting operations begin at $t = 2$ on the wildfire illustrated in Fig. 3. The fire engines are located at the purple cells. Yellow and orange cells indicate the areas controlled by the FTs and

the FH, respectively. At the next time point, these cells are completely burnt but the fire will not spread in contiguous cells further.

3.2.2. Firebreak construction

Firebreaks are the lines that serve as a barrier to slow down the progress of wildfire. A firebreak cannot catch fire as it contains no fuel. However, a fire can spread beyond the firebreak if its spread rate is large enough. A firebreak is built by removing all fuel (i.e. razing with a bulldozer) from a part of the forest land and digging down to mineral soil. Therefore, the forest land, where a firebreak is built, is damaged. However, it is reasonable to bargain away a part of forest land for protecting larger, possibly more valuable forest areas, or dwellings from a wildfire.

In Türkiye, bulldozers are used for firebreak construction. As bulldozers move slower than the fire engines, the response time for the Firebreak Construction Teams (FCTs) is longer than that of FTs. Due to the same reason, the safety distance of the FCTs from the firefront is larger than that of the FTs. Unlike the FTs, FCT's can work in front of the head fire but their safety distance from the head fire is much larger than that from the flank fire. Note that the safety distances are altered in case of a wind direction change as they are defined according

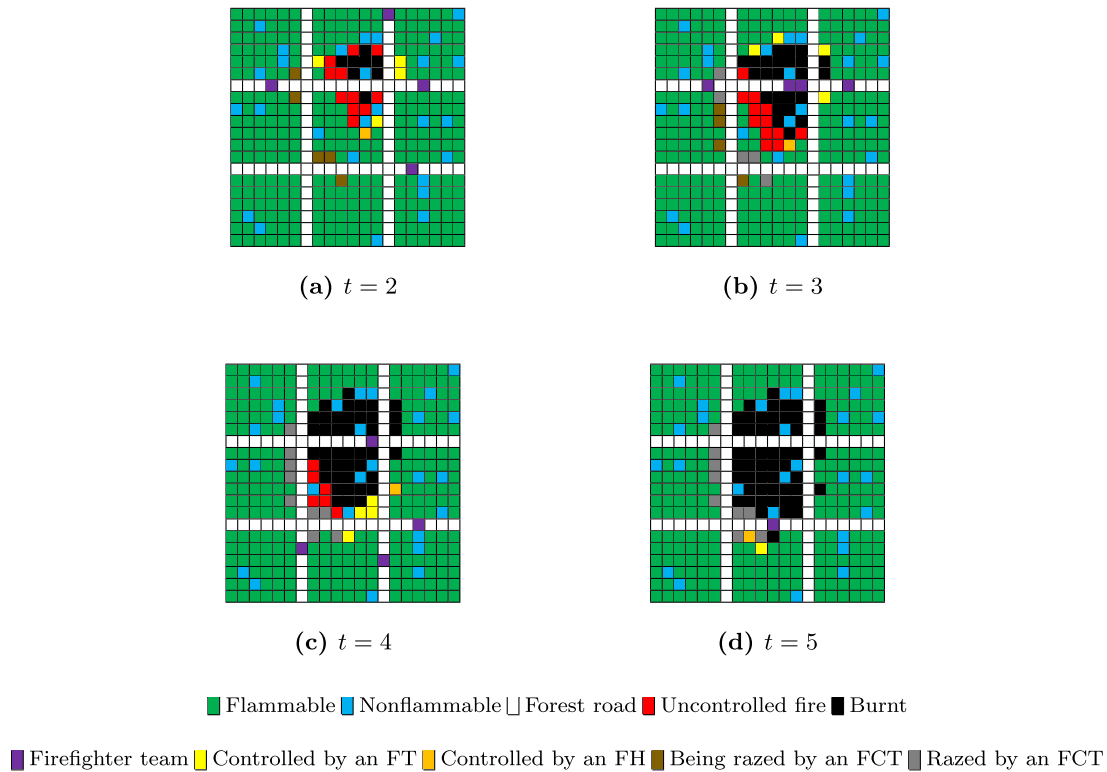


Fig. 6. An example to firebreak construction.

to the wind direction. The bulldozers can travel on the forest roads and firebreaks, therefore, an FCT can only operate on the forest areas next to a forest road or a firebreak. Accordingly, we assume that an FCT can only raze the flammable cells which are adjacent to a forest road cell or a previously razed cell.

Fig. 6 provides an example for our modelling approach for firebreak construction. In this example, we consider the same case illustrated in Fig. 3. We assume that five FCTs are available and an FCT can raze one cell in a time period. The safety distance from the head and flank fires are five and two cells, respectively. Suppose that the FCTs begin their operations in $t = 2$. In Fig. 6, the cells in which the FCTs start working are shown in grey. In the presence of FCTs, the wildfire is suppressed by $t = 6$, in shorter time and with less total damage compared to the case without FCTs.

The coordination of resources can be seen in Fig. 6. In $t = 2$ (see Fig. 6(a)), an FCT could raze a cell near the left flank of the fire thanks to an FT controlling a nearby cell on fire. The FTs working on the right flank of the fire benefited from the nearby forest road and nonflammable cells such that the controlled cells and the cells with no fuel would form a firebreak in the next period. One can observe that the FH has attacked head fire while the FTs worked on the cells in the flanks due to their safety constraints. A similar case can be seen in $t = 3$ (Fig. 6(b)). In this example, the FTs have generally worked on the right side of the fire, leaving the left side to the FCTs. One can observe that the firefighting on the right side has distorted the shape of the fire by forming a natural firebreak by extinguishing. In particular, most of the firebreak construction efforts are in earlier time periods to obtain a firebreak as soon as possible. The fire spread on the left side has been mostly stopped by this firebreak. As a result of the coordination of firebreak construction and firefighting, the wildfire has been suppressed more quickly, and the total damage of the wildfire is lower.

3.3. The complexity of the WSP

Proposition 1. *The WSP is $\mathcal{NP}$ -Hard.*

Proof. Transformation from the Knapsack Problem (KP). Consider a generic instance of the KP with n items, item sizes a_i , item values b_i , and capacity K . For the sake of simplicity, we define $A = \sum_{i=1}^n a_i$. Let us construct an instance of the WSP on a grid of seven rows and $A+n-1$ columns. Cells in columns $a_1 + 1, a_1 + a_2 + 2, \dots, \sum_{i=1}^{n-1} a_i + n - 1$ are forest roads. Cells in the first three rows that are not in the specified columns are flammable with value 1. All cells in the fourth row are forest roads. Cells in the fifth row are identical to rows one to three. Cells in the sixth row are nonflammable cells. Cells in the seventh row are flammable with value b_i/a_i for the cells between columns $i + \sum_{j=1}^{i-1} a_j$ and $i - 1 + \sum_{j=1}^i a_j$, except for $i = 1$ and $i = n$, where the former and latter boundaries are defined by the size of the grid. The rest of the cells in Row 7 are nonflammable cells.

All cells in the top row are on uncontrolled fire at $t = 0$. There is a northerly wind that enforces a safety distance of 2 cells to the south. In addition, the spread rate of the wildfire is 2 cells to the south and 1 cell in the other directions. The length of the planning horizon is computed as $t_{max} = 2 + \max_{i=1, \dots, n} a_i$. We have K FCTs from $t = 0$, with no FT or FH available. The penalty coefficient for razing is $\rho = 1$. An example of the transformation for an instance of the KP with $a_1 = 2, a_2 = 3, a_3 = 4, b_1 = 5, b_2 = 6, b_3 = 7, K = 5$ is given in Fig. 7.

By construction, FCTs can only work on cells in Row 4. The optimal solution for the example is to raze the cells corresponding to the first two columns as well as Columns 4, 5, and 6. The cells in Row 7 cannot be razed due to the lack of forest road access. Partially razing a set of cells under a contiguous set of forest cells will not inhibit the fire from spreading to the bottom row, all of which will burn due to the length of the planning horizon and the lack of forest road access. Hence the solution will contain a set of unburnt cells in Row 7, corresponding to the items selected to be in the knapsack. $\square$

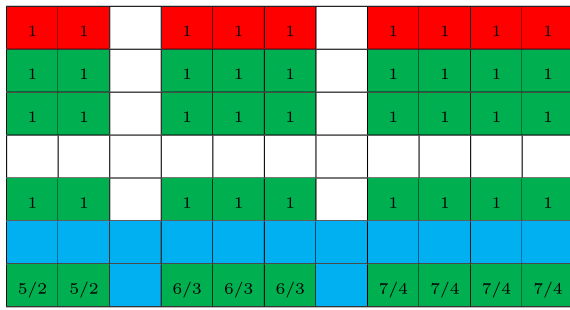


Fig. 7. An example of the transformation.

4. Mathematical formulations

We now present the notation and two mathematical formulations for the WSP.

4.1. Notation

Sets

T planning horizon, indexed by $t \in \{0, \dots, t_{max}\}$

T_g set of time points at which an FCT can start razing a flammable cell, $\{c_g, \dots, t_{max} - 1\}$, where c_g is the response delay of FCTs

T_h set of time points at which an FT or an FH can start extinguishing a fire in a flammable cell, $\{c_h, \dots, t_{max} - 1\}$, where c_h is the response delay of FTs and FHs

C set of all cells

C_r set of cells that belong to forest roads, $C_r \subseteq C$

C_f set of flammable cells, $C_f \subseteq C$

C_q set of flammable cells that are adjacent to forest roads

C_i^a set of flammable cells that are adjacent to cell $i \in C$

C_{it}^b set of flammable cells into which a fire from cell $i \in C_f$ at time $t - 1$, $t \in T \setminus \{0\}$, can spread at time t

$C_{it}^{b'}$ set of flammable cells from which a fire at $t - 1$, $t \in T \setminus \{0\}$, can spread into cell $i \in C_f$ at time t

C_i^h set of flammable cells within the extinguishing range of an FT located in cell $i \in C_r$

$C_i^{h'}$ set of road cells from which an FT can control a fire in $i \in C_f$, $C_i^{h'} \subseteq C_r$

C_{it}^{sg} set of cells in which firebreak construction cannot be done from time t to $t + 1$, $t \in T_g$, if a fire exists in $i \in C_f$ at time t

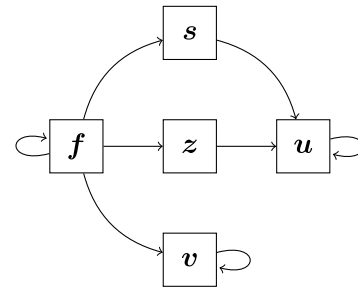
C_{it}^{sh} set of road cells in which an FT cannot be located from time t to $t + 1$, $t \in T_h$, if a fire exists in $i \in C_f$ at time t

Parameters

α_t number of FCTs available from time t to time $t + 1$, $t \in T_g$

β_t number of FTs available from time t to time $t + 1$, $t \in T_h$

γ_t number of FHs available from time t to time $t + 1$, $t \in T_h$

Fig. 8. A diagram illustrating state transitions of a flammable cell (f : initial state, s : controlled fire, z : uncontrolled fire, u : burnt, v : razed).

r_α number of cells that can be razed by an FCT in a time period

r_β number of cells that can be controlled by an FT in a time period

r_γ number of cells that can be controlled by an FH in a time period

d_i value of cell $i \in C_f$

z_i^0 1 if a fire is ignited in cell $i \in C_f$, 0 otherwise

ρ penalty coefficient for the cells razed by the FCTs

Decision variables

x_i^t 1 if an FCT works on razing cell $i \in C_f$ from time t to time $t + 1$, $t \in T_g$, 0 otherwise

y_i^t 1 if an FT is located in cell $i \in C_r$ from t to $t + 1$, $t \in T_h$, 0 otherwise

w_{ij}^t 1 if an FT is located in cell $i \in C_r$ and works on extinguishing a fire in cell $j \in C_h^i$ from t to $t + 1$, $t \in T_h$, 0 otherwise

p_i^t 1 if an FH works on extinguishing a fire in $i \in C_f$ from time t to time $t + 1$, $t \in T_h$, 0 otherwise

z_i^t 1 if an uncontrolled fire exists in cell $i \in C_f$ at time $t \in T$, 0 otherwise

s_i^t 1 if the spread of a fire in cell $i \in C_f$ at time $t \in T$ is intercepted by an FT or an FH, 0 otherwise

u_i^t 1 if the state of cell $i \in C_f$ is burnt at time $t \in T$, 0 otherwise

v_i^t 1 if the state of cell $i \in C_f$ is razed at $t \in T$, 0 otherwise

Fig. 8 shows the possible states of a flammable cell, and how the cell may transition from one to another. In the figure, each node represents a state and each arc shows a possible state transition. Node f represents the initial state which does not change if a cell neither catches fire or is razed by an FCT. Nodes s and z correspond to a controlled and an uncontrolled fire in a cell, respectively. A fire is deemed to be controlled if its spread is intercepted by either an FT or an FH, otherwise it is considered to be uncontrolled. Node u represents the state of being burnt for a cell. A cell that is in state s or z at one time point will be in state u at the subsequent time points. Finally, the state of being razed for a cell is denoted by node v . States u and v are absorbing, i.e. a burnt or razed cell does not transition to another state.

4.2. Compact formulation

Based on the notation provided above, our first formulation is:

(M_{WSP})

$$\text{Minimise } \sum_{i \in C_f} (\sum_{t \in T} d_i(z_i^t + s_i^t) + \rho \sum_{t \in T_g} d_i x_i^t) \quad (1)$$

Subject to

$$z_i^0 = z_i^0 \quad \forall i \in C_f \quad (2)$$

$$u_i^t + z_i^t + s_i^t + v_i^t \leq 1 \quad \forall i \in C_f, t \in T \quad (3)$$

$$x_j^t + p_j^t + \sum_{i \in C_f^h} w_{ij}^t \leq 1 \quad \forall j \in C_f, t \in T_g \cap T_h \quad (4)$$

$$z_j^t + s_j^t + u_j^t + v_j^t \geq z_i^{t-1} \quad \forall i \in C_f, t \in T \setminus \{0\}, j \in C_{it}^b \quad (5)$$

$$s_i^t + z_i^t \leq \sum_{j \in C_{it}^b} z_j^{t-1} \quad \forall i \in C_f, t \in T \setminus \{0\} \quad (6)$$

$$u_i^t = \sum_{t'=0}^{t-1} (z_i^{t'} + s_i^{t'}) \quad \forall i \in C_f, t \in T \setminus \{0\} \quad (7)$$

$$v_i^t = \sum_{t'=c_g}^{t-1} x_i^{t'} \quad \forall i \in C_f, t \in T \setminus \{0, \dots, c_g\} \quad (8)$$

$$s_j^t = \sum_{i \in C_f^h} w_{ij}^t + p_j^t \quad \forall j \in C_f, t \in T_h \quad (9)$$

$$v_i^t = 0 \quad \forall i \in C_f, t \in \{0, \dots, c_g\} \quad (10)$$

$$s_i^t = 0 \quad \forall i \in C_f, t \in T \setminus T_h \quad (11)$$

$$\sum_{i \in C_f} x_i^t \leq \alpha_t r_\alpha \quad \forall t \in T_g \quad (12)$$

$$\sum_{i \in C_r} y_i^t \leq \beta_t \quad \forall t \in T_h \quad (13)$$

$$\sum_{j \in C_{it}^h} w_{ij}^t \leq r_\beta y_i^t \quad \forall i \in C_r, t \in T_h \quad (14)$$

$$\sum_{i \in C_f} p_i^t \leq \gamma_t r_\gamma \quad \forall t \in T_h \quad (15)$$

$$x_i^t = \sum_{j \in C_{it}^g} v_j^t \quad \forall i \in C_f \setminus C_q, t \in T_g \quad (16)$$

$$x_j^t \leq 1 - (z_i^t + s_i^t) \quad \forall i \in C_f, t \in T_g, j \in C_{it}^{sg} \quad (17)$$

$$y_j^t \leq 1 - (z_i^t + s_i^t) \quad \forall i \in C_f, t \in T_h, j \in C_{it}^{sh} \quad (18)$$

$$y_i^t \in \{0, 1\} \quad \forall i \in C_r, t \in T_h \quad (19)$$

$$s_i^t, z_i^t, u_i^t, v_i^t \in \{0, 1\} \quad \forall i \in C_f, t \in T \quad (20)$$

$$x_i^t \in \{0, 1\} \quad \forall i \in C_f, t \in T_g \quad (21)$$

$$w_{ij}^t \in \{0, 1\} \quad \forall i \in C_r, j \in C_f, t \in T_h \quad (22)$$

$$p_i^t \in \{0, 1\} \quad \forall i \in C_f, t \in T_h \quad (23)$$

Objective function (1) minimises the total damage sustained by the fire and razing. Note that a cell on (controlled or uncontrolled) fire at a time point will be at the burnt state at the subsequent time points. Similarly, a cell razed by an FCT at a time point will be at the razed state at the subsequent time points (see Fig. 8). Constraint set (2) sets the wildfire ignition points. Constraint set (3) guarantees that at most one of the state variables can be active for each flammable cell at each time point. Constraint set (4) imposes that at most one fire suppression activity can be done in a flammable cell in a time period. Constraint set (5) dictates that if an uncontrolled fire exists in $i \in C_f$ at time $t-1$, then one of the state variables for each cell $j \in C_{it}^b$ must be active at time t . Conversely, constraint set (6) ensures that neither of the state variables z and s of cell $i \in C_f$ at time t can be 1 unless there exists

an uncontrolled fire in at least one of the cells in C_{it}^b at time $t-1$. Constraint set (7) sets the state of a flammable cell at time t to burn if a fire exists in the same cell between time points 0 and $t-1$. Similarly, constraint set (8) forces each state variable v_i^t to be 1 if cell i is razed by an FCT between time points c_g and t . Constraint set (9) ensures that a fire in $i \in C_f$ at time t cannot be under control without extinguishing it between times t and $t+1$. Since the firebreak construction starts as of time c_g , constraint set (10) imposes that the state variable v must be 0 for each flammable cell from time 0 to c_g . Likewise, due to the delay of extinguishing activities, constraint set (11) forces the state variable s to be 0 for each flammable cell from time 0 to c_h-1 . In addition, as the last possible time period for suppression activities starts at time $t_{max}-1$, the state variable s for each flammable cell should be 0 at time t_{max} . Constraint sets (12)–(15) are the resource capacity constraints. Constraint set (16) imposes that an FCT can start razing a cell only if at least one of its adjacent cells is razed or a part of the forest roads. Safety distance constraints for the FCTs and FTs are defined in constraint sets (17) and (18), respectively. Finally, constraint sets (19)–(23) specify the domains of the decision variables.

For the instances where the wind direction and the number of FCTs are constant through the planning horizon, the introduced model can be strengthened by the following valid inequality:

$$\sum_{i \in C_f} x_i^{t+1} \leq \sum_{i \in C_f} x_i^t \quad \forall t \in T_g \setminus \{t_{max}-1\}. \quad (24)$$

Inequality set (24) dictates that the number of cells razed by the FCTs in a period is nonincreasing over time, effectively cutting off symmetrical optimal solutions in which the razing activities intensify later in the planning horizon.

It is also possible to apply preprocessing to decrease the number of decision variables. We can compute the earliest time point at which the fire can spread to flammable cell $j \in C_f$, and denote it as e_j . Using this information, the variables z_j^t and s_j^t (resp. w_{ij}^t and p_j^t) can be set to 0 for each $j \in C_f, t \in T : t < e_j$ (resp. $j \in C_f, i \in C_{ij}^h, t \in T_h : t < e_j$). In addition, the variable x_j^t can be set to 0 for each $j \in C_f, t \in T_g : e_j > t_{max}$.

Alternative objective functions can easily be incorporated to the mathematical model. When the objective is the containment of the fire as fast as possible, we can set the values of all the flammable cells to be identical and remove the penalty term for the damage sustained by razing ($\rho \sum_{i \in T_g} d_i x_i^t$) from the objective function (1). By these changes, the objective function turns into the minimisation of the number of cells catching fire. Since a cell can be on fire at most for one time unit, this small modification will ensure the containment objective. Alternatively, we can introduce a variable $\bar{t}$, which denotes the last time a flammable cell catches fire. The objective function would then be the minimisation of $\bar{t}$, and the following constraint set guarantees that $\bar{t}$ is correctly set:

$$\bar{t} \geq \sum_{i \in T} t z_i^t, \quad \forall i \in C_f. \quad (25)$$

Since Muğla Regional Directorate of Forestry suggested the objective function given in Eq. (1), we employ it in the remainder of the paper.

4.3. Formulation with an exponential number of feasibility constraints

The main disadvantage of the presented M_{WSP} formulation is the complexity induced by the decision variables employed for the extinguishing operations (y , w , and p variables). Because a fire in a cell can be controlled by either the FTs located in different cells or the FHs, there exist multiple feasible resource assignments for a set of flammable cells which is determined to be controlled within a time period. Therefore, as the problem size increases, the time needed by the formulation to close the optimality gap increases considerably. In order to deal with this issue, we reformulate the WSP by projecting the y , w , and p variables out of M_{WSP} . The resulting formulation is:

$M_{WSP-R}(Initial)$: Objective function (1)

subject to

Constraints (2), (3), (5)–(8), (10)–(12), (16), (17), (20), (21),

$$\sum_{i \in C_f} s_i^t \leq \gamma_t r_\gamma + \beta_t r_\beta \quad \forall t \in T_h \quad (26)$$

Constraint set (26) limits the maximum number of cells that can be controlled in each time period. Given an integer solution $(\hat{z}, \hat{s}, \hat{x}, \hat{u}, \hat{v})$, let $I_0^t = \{i \in C_f : \hat{s}_i^t = 0\}$ (resp. $I_1^t = \{i \in C_f : \hat{s}_i^t = 1\}$) be the set of flammable cells for which the s variables are equal to 0 (resp. 1) at time $t \in T_h$. We also let $I_f^t(\hat{s}^t, \hat{z}^t)$ be the feasibility indicator function which returns 1 if the extinguishing plan between times t and $t+1$ is feasible, and 0 otherwise. Based on these definitions, the following set of combinatorial Benders' cuts (Codato & Fischetti, 2006) can be used to separate an integer infeasible solution:

$$\sum_{i \in I_0^t} s_i^t + \sum_{i \in I_1^t} (1 - s_i^t) \geq 1 \quad \forall (\hat{z}, \hat{s}, \hat{x}, \hat{u}, \hat{v}), t \in T_h : I_f^t(\hat{s}^t, \hat{z}^t) = 0 \quad (27)$$

The complete reformulation (M_{WSP-R}) is obtained by augmenting $M_{WSP-R}(Initial)$ with the feasibility cuts (27) and takes the following form:

M_{WSP-R} : Objective function (1)

subject to

Constraints (2), (3), (5)–(8), (10)–(12), (16), (17), (20), (21), (26), and (27).

In the beginning of the algorithm, the constraints (27) are dropped from the formulation and are added dynamically if they are violated by an integer solution. The separation procedure used to detect the violated constraints is explained in the next subsection.

4.3.1. Separation procedure for combinatorial benders' cuts

The following mathematical model seeks to find a feasible resource assignment plan for each time period in which the extinguishing operations are performed. The formulation uses y , w , and p variables, but the time index t is dropped. Given an integer solution $(\hat{z}, \hat{s}, \hat{x}, \hat{u}, \hat{v})$, for each $t \in T_h : I_1^t \neq \emptyset$, the resulting feasibility problem is formulated for a fixed t' value as follows:

$(M_F^{t'})$

Minimise 0 (28)

Subject to

$$\sum_{i \in C_f^{t'}} w_{ij} + p_j = 1 \quad \forall j \in C_f : \hat{s}_j^{t'} = 1 \quad (29)$$

$$\sum_{i \in C_r} y_i \leq \beta_t \quad (30)$$

$$\sum_{j \in C_h^{t'} : \hat{s}_j^{t'} = 1} w_{ij} \leq r_\beta y_i \quad \forall i \in C_r \quad (31)$$

$$\sum_{i \in C_f : \hat{s}_i^{t'} = 1} p_i \leq \gamma_t r_\gamma \quad (32)$$

$$y_j = 0 \quad \forall j \in C_{i,t'}^{sh} : i \in C_f, \hat{z}_j^{t'} + \hat{s}_j^{t'} = 1, \quad (33)$$

$$y_i \in \{0, 1\} \quad \forall i \in C_r \quad (34)$$

$$w_{ij} \in \{0, 1\} \quad \forall i \in C_r, j \in C_f : \hat{s}_j^{t'} = 1 \quad (35)$$

$$p_i \in \{0, 1\} \quad \forall i \in C_f : \hat{s}_i^{t'} = 1. \quad (36)$$

The objective function (28) is empty since $M_F^{t'}$ is a feasibility problem. Constraints (29)–(36) are respectively the counterparts of the sets (9), (13)–(15), (18), (19), (22), and (23) from the M_{WSP} formulation. The solution $(\hat{z}, \hat{s}, \hat{x}, \hat{u}, \hat{v})$ is feasible to the original problem if M_F returns a feasible solution for each of the time period examined. Otherwise, if it returns an infeasibility for at least one time period, the feasibility cuts (27) are added to the formulation. Note that the inequality set (24) and the preprocessing for the M_{WSP} formulation are also valid for the M_{WSP-R} formulation.

5. Computational analysis

This section presents a computational study performed on a set of randomly generated WSP instances. The set contains three groups of instances, with the number of cells being 25×25 (small-size), 50×50 (medium-size), and 100×100 (large-size). The forest roads were designed to be evenly distributed both horizontally and vertically. For an instance with $m \times n$ cells, the number of horizontal and vertical forest roads are determined as $\lceil m/10 \rceil$ and $\lceil n/10 \rceil$, respectively. Of the cells not belonging to forest roads, 10% of them were randomly selected to be non-flammable. The forest road cells and non-flammable cells have no value in terms of the objective function. The value of each flammable cell is generated by first selecting a random number r from the interval $(0, 10]$ and then rounding it to the smallest integer that is greater than or equal to r . The penalty coefficient ρ for razing a cell was set to 0.1.

The head fire, flank fire, and back fire spread rates are assumed to be identical for each flammable cell and determined as 4, 2, and 1 cells per time unit, respectively. The fire ignition points were randomly selected from the flammable cells. For the small-size instances, a single ignition point was considered. For the medium-size and large-size instances, the number of ignition points are $\{1, 2\}$ and $\{1, 2, 3\}$, respectively. For each instance, the length of the planning horizon is 10 and the wind direction is from north to south for each time period. Note that, as the sizes of our formulations depend on the planning horizon length, we conducted preliminary experiments to appropriately set it in terms of both fire suppression requirements and computational burden. The response delay is set to 2 time units for the FTs and FHs, and 3 for the FCTs.

To specify the levels of resources which are assumed to be constant over time, we examined Wildfire Suppression Action Plan of Türkiye (Orman Genel Müdürlüğü, 2021). We have observed that the number of FTs is always more than the number of FCTs and the number of FHs is always the least among all. Based on this observation, the sets of values for the number of FTs, FCTs, and FHs were determined as $\{5, 6, 7\}$, $\{2, 3, 4\}$, and $\{1, 2\}$, respectively. As a result, we obtained $3 \times 3 \times 2 = 18$ different resource configurations. We generated one instance for each combination. Consequently, the small-size, medium-size, and large-size data sets contain 18, 36, and 54 instances, respectively. For the instances with multiple ignition points, the resource levels were multiplied by the number of ignition points. This is in line with the real-life cases because when more than one fire occurs in a region, the suppression resources from neighbouring regions can be sent to the relevant region. The data is provided in the online supplement.

The formulations were coded in C++ and solved using Gurobi 10.0 (Gurobi Optimization, 2022). The lazy callback routine of Gurobi has been utilised to add the feasibility cuts (27) dynamically. The computational experiments were conducted on a personal computer with 4.00 GHz Intel Core i5-760X processor and 32 GB RAM. In all executions, we imposed a computational time limit of 3600 s. The results of the computational experiments are presented in Tables 2–4 and Fig. 9. The detailed computational results are presented in the online supplement.

The rest of this section is organised as follows: In Section 5.1, the performance of the proposed approach is analysed. Section 5.2 presents the results obtained by considering a wind direction change in the middle of the planning horizon. In Section 5.3, a sensitivity analysis on the response delay of suppression resources is presented. Finally, Section 5.4 investigates the effect of the forest road network structure on the success of wildfire suppression.

5.1. The performances of WSP formulations

In this section, the performances of the formulations are tested on the generated WSP instances. The average size information for each formulation and instance group is reported in Table 1, where

Table 1

Average size information for each formulation and instance group.

Ins. group	M_{WSP}			M_{WSP-R}				M_F	
	#V	#C	#EV	#V	#C	#EV	#CBC	#V	#C
Small-size	37 476	156 583	34 438	12 869	136 341	8892	46	243	260
Medium-size	82 768	744 806	195 858	40 753	651 693	50 482	53	626	778
Large-size	196 373	3 237 241	887 068	133 045	2 715 101	231 548	115	2075	2705
All	132 022	1 892 986	514 560	82 252	1 597 505	134 083	83	1287	1655

Table 2

Average results obtained by the formulations.

Ins. group	M_{WSP}				M_{WSP-R}			
	Gap%	t (s)	#O	#NO	Gap%	t (s)	#O	#NO
Small-size	0.0	45	18	0	0.0	30	18	0
Medium-size	0.0	141	36	0	0.0	98	36	0
Large-size	2.1	1147	44	10	0.2	675	53	1
All	1.1	628	98	10	0.1	375	107	1

#V, #C, #EV, and #CBC represent the average numbers of variables, constraints, eliminated variables by preprocessing, and combinatorial Benders' cuts, respectively. The table shows that by applying the preprocessing, the number of decision variables in M_{WSP-R} and M_{WSP} are reduced by an average of 62.0% and 79.6%, respectively. Moreover, it is seen that the average numbers of decision variables and constraints of M_{WSP-R} are about 37.7% and 15.6% less than those of M_{WSP} , respectively.

Table 2 presents the average results obtained by M_{WSP} and M_{WSP-R} for each instance group. Gap% denotes the average optimality gap percentage. t (s) stands for the average CPU time in seconds. While #O denotes the number of instances solved to optimality, #NO is the number of instances for which the optimality has not been proven within the specified CPU time limit.

The results indicate that M_{WSP-R} outperform M_{WSP} significantly. It has been observed that M_{WSP-R} yields the best average lower and upper bound values for each instance group with a much lower average CPU time. Specifically, the superiority of M_{WSP-R} over M_{WSP} is more apparent in large-size instances, for which M_{WSP-R} is capable of solving 53 out of 54 instances to optimality, significantly more than M_{WSP} that can only solve 44. Although a computational time limit of one hour was imposed for each execution, M_{WSP-R} is able to solve most of the instances to optimality in much shorter computing times. For example, when the time limit has been set to 5, 10, and 15 min, M_{WSP-R} has still managed to prove optimality in respectively 79, 90, and 96 out of 108 instances.

We have observed that the formulations become more challenging to solve when the resource levels are low. Accordingly, the computational times for M_{WSP-R} obtained for the instances with the lowest and highest resource levels are respectively averaged around 678.3 and 40.4 seconds. Furthermore, in line with this observation, the only instance for which the optimality cannot be proven by M_{WSP-R} belong to the instance group with the lowest firefighting resource level.

5.2. Analysing the effect of a change in the wind direction

In order to analyse the effect of a wind direction change, we have considered a scenario in which the wind direction changes by 90 degrees in the middle of the planning horizon. In this analysis, we assume that the time of the wind change and the new wind direction are known at the beginning of the planning horizon. Particularly, the analysis was performed on the large-size WSP instances by considering that the wind shifts to the east-to-west at $t = 5$. Table 3 presents the average results obtained for each scenario, where all the instances have been solved by M_{WSP-R} . In this table, the instances with the same resource levels are grouped together. The column “#Res” corresponds to the sum of the numbers of FTs, FHs, and FCTs per ignition point.

Table 3

Average results of the analysis of a wind direction change.

#Res	#Ins	Without wind direction change		With wind direction change		Difference (%)	
		UB	t (s)	UB	t (s)	UB	t (s)
8	3	1395.6	1132	1339.5	1202	4.0	-6
9	9	1178.6	1545	1047.5	369	11.1	76
10	15	1010.6	641	975.5	510	3.5	20
11	15	833.8	478	820.6	446	1.6	7
12	9	778.8	239	765.8	113	1.7	53
13	3	664.7	69	664.7	76	0.0	-10
All	54	953.0	675	912.5	417	4.3	38

The column “#Ins” represents the number of instances in each instance group. Columns “UB” and “ t (s)” denote the average upper bound values and CPU times, respectively. The column “Difference (%)” shows the average percentage differences between the results obtained for the scenarios with the wind direction change and the base scenarios.

The results indicate that the average objective function values of the solutions obtained under the wind direction change scenario are generally lower than those obtained under the base case scenario. This is an expected outcome since our model is deterministic and we have a complete information on when and in which direction the wind turns. With this knowledge, the FTs and FHs attack to the west side of the firefront. When the wind direction changes from the north to the east, the west side of the firefront is already under control. The controlled cells in the west serve as a firebreak to dampen the head fire spreading from the east. As a result, the suppression efforts become more effective, and the wildfire is suppressed more quickly. Consequently, the average value lost is lower in the wind direction change scenario. As has been experienced for the objective function values, it has been observed in most of the instances that the computing time on the wind-turned scenario is lower than that on the base-case scenario. The results of this analysis highlights the importance of accurate weather forecast in wildfire suppression. Although predicting the wind direction and speed at certainty is not possible due to the chaotic nature of the weather, advanced forecasting techniques can provide very precise estimates (Wang et al., 2021). In practice, the wildfire professionals take advantage of wind change forecasts in their decisions. In this regard, Nuraiman et al. (2022) incorporated the wind direction change forecast into their model to effectively deploy the resources.

5.3. Analysing the effect of response delay on the success of fire suppression

In the related literature, the impact of response delay on the fire containment performance of initial attack teams was investigated by Ntairo et al. (2013), Rashidi et al. (2018b), Wei et al. (2015). Here, our aim is to analyse the impacts of the response delays of different types of resources on the success of wildfire suppression. Accordingly, we have conducted a numerical study on the medium-size WSP instances. The results are depicted in Fig. 9. We have considered four levels for the response delays of firefighting and firebreak construction resources. Firstly, we examined the effect of the response delay of FCTs. In this analysis, the response delay of firefighting is set to 2. The

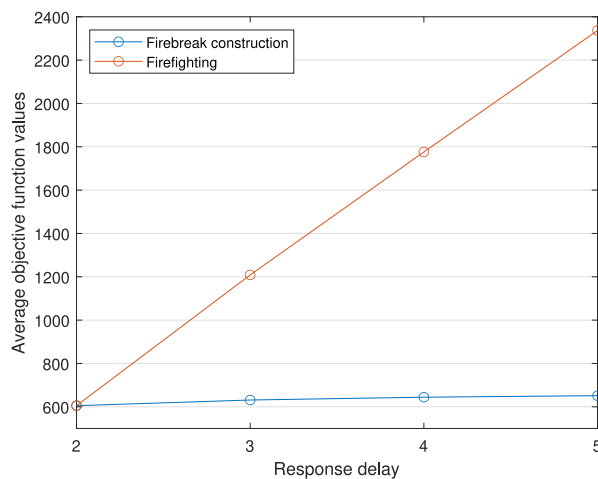


Fig. 9. The effect of changes in response delays on the total value lost.

average objective function value increases by 7.2% when the response time of FCTs increases from 2 to 5.

In the second analysis, we have investigated four levels of firefighting response delay. Similar to the first experiment, the response delay of FCTs remains constant at 2. As can be seen from 9, the response delay of firefighting is much more critical than that of firebreak construction. The change in the average objective function value is almost linear with respect to the firefighting response delay. The average total lost value increases 3.9 times when the response delay of firefighting rises from 2 to 5.

5.4. Analysing the impact of forest road network structure on the fire suppression

The success of wildfire suppression depends on the structure of the forest road network or firebreaks. The ground suppression resources access the firefront by travelling on forest roads. In addition, forest roads serve as an anchor point in ground firefighting and firebreak construction. Particularly, the FTs are deployed on the forest roads and attack the firefront by laying hoses into the forest land. Likewise, the FCTs can work on the zones which are adjacent to the forest roads. Moreover, forest roads constitute fuel breaks on the forest land. We refer the readers interested in the details of fuel break construction to León et al. (2019), Recasens et al. (2022), Rytwinski and Crowe (2010), Wei (2012). The forest road network should be sufficiently dense to ensure the effectiveness of wildfire suppression (Laschi et al., 2019; Thompson et al., 2021). However, a considerable amount of the forest land is lost at the expense of building the forest roads (Demir, 2007; Heinemann, 2017). Hence, there is a trade-off between the effectiveness of wildfire suppression and the value lost during the road construction. In this context, we have performed a computational experiment to analyse the effect of forest road network structure on the effectiveness of wildfire suppression by varying the level of resources available to suppress the wildfire. We have generated an additional set of small-size instances by considering two alternative forest road network structures. These instances involve only flammable cells with the same value for each cell. We considered two forest road network structures, namely, N-I and N-II, which are illustrated in Fig. 10. Both structures involve two vertical forest roads, however, N-II involves one additional horizontal road. We set nine alternative ignition points which are marked as red in Fig. 10. The ignition points are the same for both network structures. The combinations of resource levels are the same as in the generated WSP instances. We generated one instance for each ignition point and resource combination. As a result, 324 additional instances were generated to perform this analysis.

Table 4

Average results obtained for the forest road network structures.

#Res	#Ins	N-I	Without penalty		With penalty	
			N-II	Diff%	N-II	Diff%
8	9	530.8	503.1	5.2	514.6	3.1
9	27	497.2	468.9	5.7	480.4	3.4
10	45	470.0	441.9	6.0	453.4	3.5
11	45	438.3	414.0	5.5	425.5	2.9
12	27	408.8	388.5	5.0	400.0	2.2
13	9	370.7	362.8	2.1	374.3	−1.0
All	162	453.4	428.8	5.4	440.3	2.9

A summary of the results are given in Table 4, where the instances are grouped according to their resource levels (#Res). The columns N-I and N-II include the average total value lost in N-I and N-II instances, respectively. Here, the results obtained for N-II instances are represented in two sections. While the section “Without penalty” presents only the average values lost during the wildfires, in the section “With penalty”, in addition to the damage caused by the wildfire, the value lost at the cost of building the additional forest road is also taken into account. In this analysis, the penalty rate for razing a cell to build a forest road is set to 0.1. The column Diff(%) shows the average differences between Columns N-I and N-II. The results show that the damage caused by the wildfires can be reduced by an average of 5.4% by building the extra forest road. When the values lost in the road construction are included, the average reduction is 2.9%. In addition, it has been observed that the benefit obtained from the extra road construction is generally higher in the instances with a low resource level than in those with a high resource level.

6. Case study

To demonstrate the applicability of our model, we present a case study of wildfire suppression in Yatağan district of Muğla province, Türkiye. Yatağan has a high wildfire risk due to its typical Mediterranean climate with hot and dry summers. Moreover, approximately 70% of Yatağan surface area is covered by Turkish pine (*pinus brutia*) which is susceptible to wildfire. Therefore, Yatağan forest is classified as a first-degree wildfire hazard zone in Türkiye. Yatağan Directorate of Forest District (DFD) is responsible for forest management and wildfire suppression in Yatağan. Currently, Yatağan DFD has six FTs, an FH with night vision goggles and two FCTs (Orman Genel Müdürlüğü, 2021). When a wildfire escapes from the initial attack, the neighbouring forest directorates act in coordination to suppress it. Therefore, Yatağan DFD can quickly be supported by resources of other directorates in case of a large wildfire. Moreover, Yatağan DFD has a right to commandeer the vehicles of nearby mining companies for firebreak construction during a wildfire.

In the case study, we consider a 5.504 km × 4.936 km region from a forest map provided by Yatağan DFD. A satellite image of the considered region is given in Fig. 11(a). We rasterised the region into 183 × 164 cells each of which has 0.03 km × 0.03 km size. The forest map provided by Yatağan DFD includes detailed information about forest flora, geographical formations, forest roads, infrastructure, and populated zones. We consider different fire spread rates for the cells depending on their fuel type. By consulting the managers, we have specified the fire spread rates as in Table 5. The highest fire spread rate is associated with the closed canopy Turkish pine and mixed forests while the spread rate on agricultural and residential zones is the lowest. As the back fire spread rate on maquis shrubland, degraded forest, agricultural and residential zones is negligible, we assumed that back fire cannot spread on the cells with these fuel types. The cells related to streams, lakes, mining zones, forest roads, and highways are marked as nonflammable since they do not contain any fuel. It is worth noting that we do not consider the affect of slope on the fire spread since we

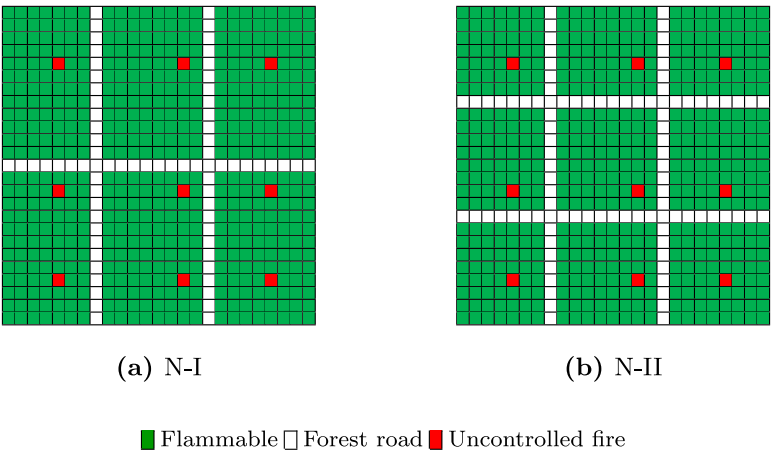


Fig. 10. Alternative forest road network structures. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

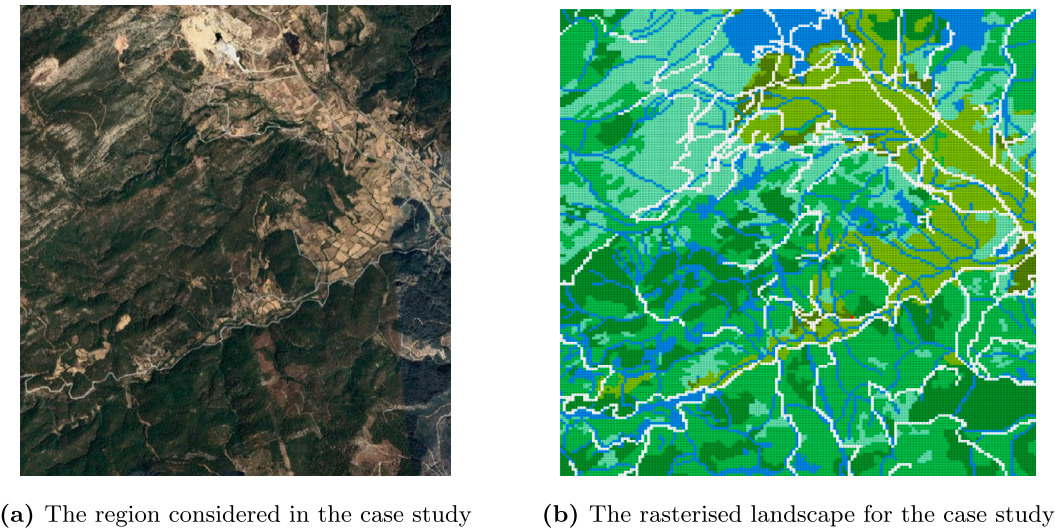


Fig. 11. The rasterised landscape for the case study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5
Fire spread rates considered in the case study.

Cell type	Spread rates					
	Head fire		Flank fire		Back fire	
	(km/h)	(cell/h)	(km/h)	(cell/h)	(km/h)	(cell/h)
Closed canopy Turkish pine forest	0.24	8	0.12	4	0.03	1
Mixed forest	0.24	8	0.12	4	0.03	1
Open and semi-closed canopy Turkish pine forest	0.18	6	0.09	3	0.03	1
Maquis shrubland	0.12	4	0.06	2	0	0
Degraded forest	0.12	4	0.06	2	0	0
Agricultural zone	0.06	2	0.03	1	0	0
Residential zone	0.06	2	0.03	1	0	0
Stream	0	0	0	0	0	0
Natural and artificial lakes	0	0	0	0	0	0
Mountain ridge	0	0	0	0	0	0
Mining zone	0	0	0	0	0	0
Forest roads and highways	0	0	0	0	0	0

could not obtain the detailed slope data on the region. However, the cells which constitute mountain ridges are classified as nonflammable to reflect the dampening effect of mountain ridges on fire spread. The resulting rasterised landscape is illustrated in Fig. 11.

The values of cells are different depending on their types. The cells corresponding to residential zones have the highest value of 9. The values associated with agricultural zones and forests are 4 and 1, respectively. The nonflammable cells have no value as they do not

contain any fuel. Particularly, the cells associated with mining zones are marked as nonflammable and a wildfire cannot spread on a mining zone. On the other hand, it is undesirable that a fire reaches a mining zone. Therefore, we set the highest value, 9, to the flammable cells which are adjacent to the mining zone.

In the wildfire season, the dominant wind direction is from the north to the south (Yatağan Orman İşletme Müdürlüğü, 2020). We consider a wildfire scenario in which a cell belonging to a closed canopy forest

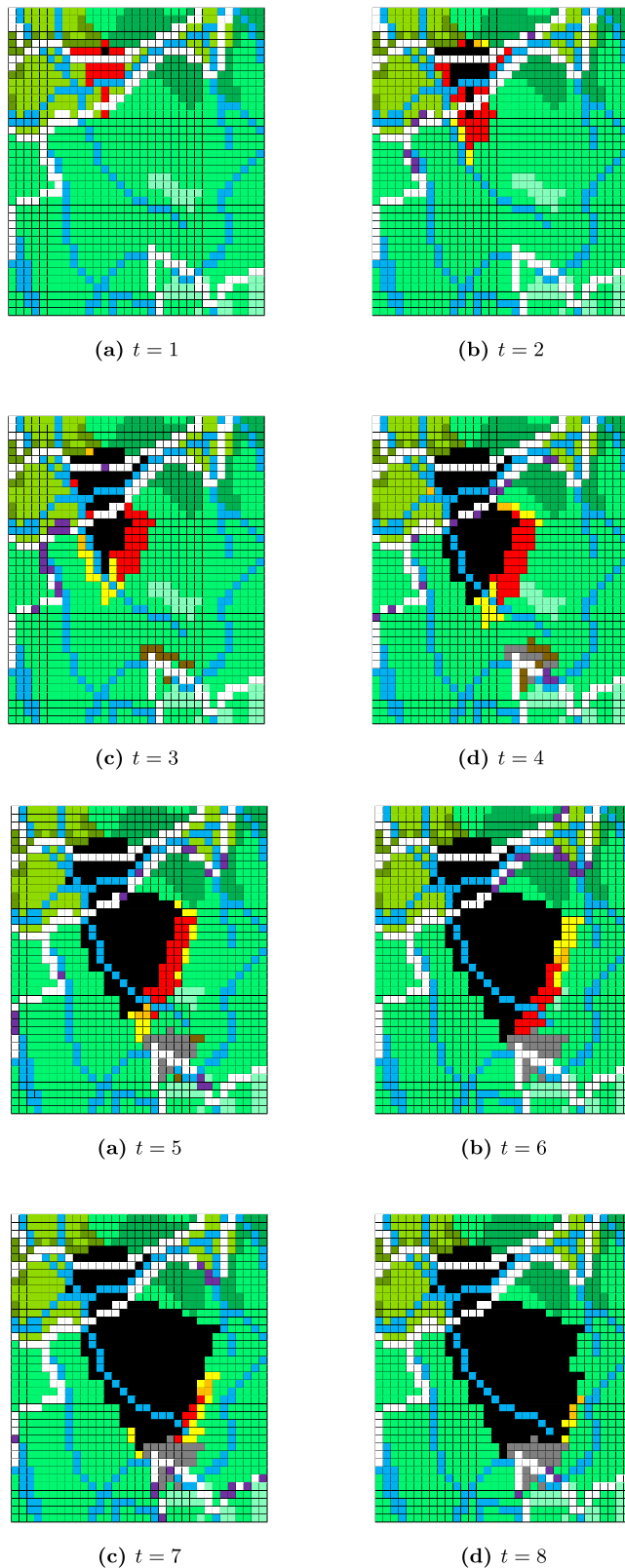


Fig. 12. The optimal solution for the case study.

catches fire from the electricity transmission line. The ignition cell is represented in Fig. 11 in red colour. In this application, a time period corresponds to one hour. Consequently, we assume that the response

Table 6

The suppression resources available at each time.

Time	Number of FTs	Number of FHs	Number of FCTs
$t \in \{0, 1\}$	None	None	None
$t = 2$	6	1	None
$t \in \{3, \dots, t_{max}\}$	14	3	10

delays for the firefighting and firebreak construction are two and three hours, respectively.

At $t = 2$, the firefighting begins with the resources of Yatağan (six FTs and an FH). At $t = 3$, two bulldozers of Yatağan DFD and eight bulldozers belonging to the nearby mining company become available for firebreak production, for a total of 10 FCTs. Moreover, we considered the additional firefighting resources provided by neighbouring directorates, Milas and Muğla DFDs. Milas and Muğla DFDs have 13 and 9 FTs, respectively. Both DFDs have one FH each (Orman Genel Müdürlüğü, 2021). We assume that Milas and Muğla DFDs send their FHs and one thirds of their FTs to Yatağan. We assume that the additional firefighter resources from both DFDs are available at $t = 3$ as the distances of Milas and Muğla to Yatağan are approximately equal. Table 6 summarises the resources available at each time point.

As a result of the preliminary experiments, the length of the planning horizon has been set to 10 h. An optimal solution to the problem has been reached in 4974.19 s. The optimal solution is illustrated in Fig. 12. Due to the space limitation, we provide the case study data and the optimal solution in the online supplement. On the north-west side of the ignition point, there are residential and agricultural zones. Since the backfire spread rate is zero in these zones, the backfire is quickly controlled. At the beginning, firefighting resources are deployed to attack west side of the wildfire while a firebreak construction begins on the south-east side. Notice that there is a creek, which is a nonflammable line, on the west side of the wildfire (See Fig. 12c). FTs and FHs attack the cells adjacent to the creek to prevent the wildfire from spreading beyond it. As a result of firefighting, the nonflammable and controlled cells form a natural firebreak. Additionally, one can observe that FTs take the advantage of burnt areas and firebreaks to conform their safety requirements. This is consistent with the safety procedures employed in practice (The Scottish Government, 2013). The wildfire is under control by $t = 8$.

In order to test the importance of timing of support from neighbouring DFDs, we also run our algorithm for two different scenarios, in each of which the additional resources from one neighbouring DFD is delayed by one hour. It has been observed that the total damage caused by the wildfire increases by 21.7% (resp. 10.7%) in the scenario where the resources from Milas (resp. Muğla) are available at $t = 4$. This outcome reveals that even just one hour delay in support from one of the neighbouring DFDs significantly increases the damage of the wildfire. In our case study, Yatağan DFD has fewer firefighting resources than other two neighbouring DFDs. To observe the effect of resource positioning, we conduct numerical experiments for another two different scenarios, in each of which the firefighting resources of Yatağan and those of a neighbouring region are swapped. The results indicate that the total damage decreases by 20.1% in the scenario involving a resource exchange with Muğla DFD. This decrease rate increases to 40.8% in the scenario with a resource swap with Milas DFD, which highlights the importance of resource positioning in wildfire suppression.

The results reveal the applicability of the proposed formulation to real-life, large-size WSPs. Moreover, the results have shown that the coordination of multiple types of resources and suppression plans tailored to forest terrain and flora have a significant impact on the success of wildfire suppression. Note that the WSP is a planning problem which allows the preparation of fire suppression plans for predefined scenarios. Forestry organisations can use the proposed formulation to

test alternative resource settings under different wildfire scenarios. In this study, the WSP is defined on a raster landscape comprised of $0.03 \text{ km} \times 0.03 \text{ km}$ size cells to boost the resolution of the solutions as much as possible. Such level of resolution may not be required in many practical implementations. In such cases, the cell size may be increased to reduce the problem size, and consequently, obtain optimal solutions in shorter computing times. For instance, Forest Management Regulation of Türkiye restricts the minimum sizes of forest compartments and sub-compartments to 50 and 3 hectares, respectively (Orman Genel Müdürlüğü, 1991). Here, $0.055 \text{ km} \times 0.055 \text{ km}$ size cells would be sufficient to conform a sub-compartment level of resolution. In that case, we could rasterise the considered region into 100×90 cells, which would result in a much smaller problem size. Furthermore, the proposed formulation may be employed to develop initial attack plans in limited computation times without a proof of optimality.

7. Conclusion

In this study, we have addressed a WSP in which multiple types of suppression resources are considered with their operational constraints and effects on wildfire spread. We propose a basic IP formulation and a reformulation with combinatorial Benders' cuts for the problem. Our formulations cover the operational characteristics and limitations of specific types of suppression resources. Additionally, it encompasses the impact of the suppression on wildfire spread with an explicit definition of distinct effects of the resources. We have tested our formulations on a set of randomly generated instances. The results indicate the effectiveness of the reformulation in solving the WSP. Moreover, we have presented a case study for the application of the proposed reformulation to a wildfire case in Yatağan district of Muğla province in Türkiye. The results obtained for the case study reveal the effectiveness of the proposed algorithm in solving complex real-life problems.

Our research is informed by interviews conducted with Muğla Regional Directorate of Forestry in Türkiye. Hence, the operational capabilities and constraints of the resources are modelled based on the ground and aerial suppression tactics and available resources in Türkiye. However, the models can be generalised to cover the characteristics of other types of resources with small modifications. The possible future extensions of this study may include different suppression tactics such as back-burning or the usage of unmanned aerial and unmanned ground vehicles. Moreover, the proposed model can be extended with routing decisions to incorporate the travelling of resources from one cell to another. Apart from these, a procedure to identify a proper upper bound on the length of the planning horizon can considerably increase the efficiency of the algorithm. Furthermore, heuristic approaches may be developed to find high quality solutions in shorter computation times.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ejor.2024.03.005>.

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